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Bioinspired dopamine and zwitterionic polymers for non-fouling surface engineering

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Biofouling is a serious problem in the medical, marine, and all other industrial fields as it poses significant health risks and financial losses. Therefore, there is a great demand for endowing surfaces with antifouling properties to mitigate biofouling. Zwitterionic polymers (containing an equimolar number of homogeneously distributed anionic and cationic groups on the polymer chains) have been used extensively as one of the best antifouling materials for surface modification. Being a superhydrophilic polymer, zwitterionic polymers need a strong binding agent to continue to remain attached to the surface for long-term applications. The use of a mussel-inspired dopamine adhesive functional layer is one of the most widely exploited approaches for the attachment of a zwitterion layer on the surface via thiol and amine chemistry. Based on recent studies, we have categorized this dopamine and zwitterion conjugation into four different approaches: (1) conjugation of dopamine with zwitterions by direct modification of zwitterions with the dopamine functional moiety; (2) co-deposition of dopamine with zwitterionic polymers; (3) zwitterionic post modification of the polydopamine (PDA) coated surface; and (4) surface-initiated polymerization of zwitterionic polymers using dopamine modified initiators. In this review, we have briefly discussed about all the possible conjugation mechanisms and reactions for this promising dopamine and zwitterion conjugation and how this conjugated system significantly contributes to the development of non-fouling surfaces along with the other applications.

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1. Introduction

1.1. Why non-fouling?

Biofouling is highly undesirable for many applications and has become a costly and serious problem worldwide in the medical,



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marine and industrial fields. Biofouling arises from non-specific interactions between wetted surfaces and foulants such as micro- and macro-organisms, biomolecules, plants or algae.¹ Prosthetic implants, biosensors, catheters, vascular stents, dental implants and other medical equipment are prone to surface biofouling due to adhesion of microbial or thrombotic agents.² Infection from these biomedical devices and implants can lead to death of the patient. Severe biofouling of ships and the underwater structure, ultimately resulting in failure, has become a common threat in the marine industry. Increased drag, corrosion, loss of speed, increased fuel consumption *etc.* are the problems resulting from marine biofouling which are also responsible for significant financial losses in this sector.³ Power plants, textile industries, water-treatment systems and food/beverage industries are the most affected industrial sectors because of biofilm formation.^{4,5} Production loss, damage to materials, pipe blockage, decreased membrane flux, contaminated water, reduced heat-exchanger efficiency *etc.* are the problems resulting from industrial fouling. The annual total cost associated with biofouling in the marine industry is nearly \$1 billion.⁶ The cost associated with infections from biofouling in the medical industry in the United States alone is \$6 billion per year.⁷ Therefore, endowing the surface with antifouling and antimicrobial properties to mitigate the biofouling is in great demand. The surface chemistry and physical properties of the material are both crucial for preventing the recruitment of biofouling organisms. Antifouling commonly refers to all systems that prevent an organism from attaching to a surface. Natural antifouling surfaces exhibit both chemical and physical attributes. Sharks, mussels, lotus leaf and crabs are common examples of marine organisms that possess natural antifouling defenses.⁶ They offer many physical and chemical mechanisms such as low drag, adhesion, wettability (water repellency and attraction) *etc.* to control fouling. Inspired from nature,

researchers are currently designing antifouling surfaces to combat biofouling, which are more environmentally friendly and nontoxic than those obtained by the previously reported approaches. Earlier, the term antifouling was associated only with the incorporation of biocidal agents such as copper, zinc, tributyltin (TBT) *etc.* on surface coatings with a gradual release system.^{6,8} The excess release of biocidal compounds is detrimental to non-target organisms and the surrounding environment. A potential alternative approach towards combating biofouling is to alter the characteristics of the material surface to a less favorable biological attachment. In this regard, polymers are considered as one of the most prospective materials because of their wide range of antifouling and antimicrobial properties with excellent biocompatibility. A broad range of antifouling strategies are currently being employed ranging from coatings to cleaning techniques. Among them, maintaining a strong hydration layer on top of the surface is believed to be the best approach to mitigate surface fouling.⁹

1.2. Zwitterion

Among all the approaches for developing novel antifouling surfaces for non-specific protein repulsion, the incorporation of hydrophilic polymer layer is the most promising one.¹⁰ Highly hydrated chemical groups with optimized physical properties of the surface, along with the appropriate surface coating methods, are the effective strategies for developing stable non-fouling materials for long-term applications. Chen *et al.* have classified these non-fouling materials into two major categories, polyhydrophilic and polyzwitterionic.¹¹ Non-fouling mechanisms of both polyhydrophilic and polyzwitterionic materials are based on creating a hydration layer near the surface and steric repulsion.^{11–13} The hydration layer formed by hydrophilic polymers such as PEG and their derivatives, is maintained by weak hydrogen bonds.¹⁴ On the other hand, the hydration layer formed by zwitterionic polymers is more tightly bound through electrostatic interactions which makes these polymers more effective in resisting the adhesion of fouling agents.¹⁵ PEG based polymers are composed of one oxygen atom in each repeat unit ($-\text{CH}_2\text{CH}_2\text{O}-$) which can bind only one water molecule through a hydrogen bond, while zwitterions can bind up to eight water molecules by forming electrostatic forces which is stronger than the hydrogen bonding in PEG/water systems.¹⁶ Moreover, reports have shown that PEG based hydrophilic polymers degrade by oxidation, especially in complex media, which limits their long-term applications.¹⁰ Zwitterionic polymers have an equimolar number of homogeneously distributed anionic and cationic groups on their polymer chains. Non-fouling zwitterionic polymers can be of two types: polybetaines with positively and negatively charged moieties on the same monomer unit and polyampholytes with positively and negatively charged moieties on different monomer units.¹¹ Polybetaines can be further classified into three major groups based on the negatively charged groups: sulfonate-betaines (SB), carboxylate-betaines (CB), and phosphonate-betaines (PB). Among them phosphobetaine-based zwitterionic polymers are considered as biomimetic



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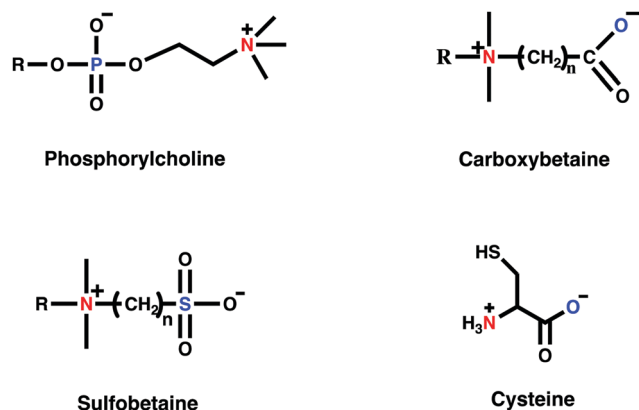


Fig. 1 Zwitterionic functional groups: phosphorylcholine, sulfobetaine, carboxybettaine, and cysteine. Redrawn with permission from ref. 21. Copyright 2014 American Chemical Society.

fouling resistant materials as they have phosphorylcholine headgroups which are usually found in the outside layer of cell membranes.¹⁷ The polyampholyte polymer is another class of antifouling polymers which are called *pseudo-zwitterionic* polymers and are composed of a pair of separate monomers with uniformly distributed two opposite charge moieties respectively.¹⁸ Polyampholytes with a perfect balance of positive and negative moieties mimic zwitterionic materials and exhibit excellent antifouling properties.^{19,20} Uniform charge distribution and charge neutrality of two closely connected opposite charge moieties on the zwitterionic materials are the main reasons for forming a maximum electrostatically induced hydration layer and avoiding favorable interaction with protein molecules.^{11,16} Because of the simplicity of synthesis, ease of applicability and availability of functional groups zwitterionic polymers are considered one of the promising non-fouling biomaterials. Some of the common zwitterionic functional groups are shown in Fig. 1.

1.3. Dopamine

Since first reported by Messersmith and co-workers in 2007,²² mussel-inspired dopamine chemistry has been increasingly used for surface modification due to its simplicity, versatility and strong reactivity for secondary functionalization.^{23,24} Messersmith and co-workers reported that under oxidative conditions dopamine with functional catechol and amino groups can be self-polymerized to form a single-step thin coating of polydopamine (PDA) on a wide range of substrates.²² This approach is mainly inspired by the wet adhesive biomolecule-foot proteins secreted by mussels for attachment to all types of inorganic and organic surfaces.^{22,25} The amino acid composition of these foot proteins that are found near the plaque-substrate interface, has two key features that inspired PDA: (i) high catechol (3,4-dihydroxybenzene) content due to the presence of 3,4-dihydroxy-L-phenylalanine (DOPA); and (ii) high primary and secondary amine content due to lysine (Lys) and histidine residues.^{26,27} Messersmith and co-workers hypothesised that the coexistence of catechol and amine functional groups is

responsible for strong adhesion to a wide spectrum of materials and identified dopamine as a small-molecule compound that contains both functionalities.²² They reported that dopamine acts as a powerful building block for the spontaneous deposition of thin polymer films of polydopamine on virtually any bulk material surface and the deposited films can easily be adapted for a wide variety of functional uses.^{22,28}

1.3.1. Polydopamine structure and formation. Understanding the chemistry of polydopamine formation is a challenging subject and the details of this mechanism still remain an active area of investigation. A consensus in polydopamine chemistry begins with the spontaneous deprotonation of hydroxyl groups under oxidative condition and formation of dopamine-quinone.²⁹ This oxidation product dopamine-quinone undergoes nucleophilic intramolecular cyclization and later turns into leukodopaminechrome which subsequently forms 5,6 dihydroxyindole or 5,6 indolequinone through further oxidation and rearrangement.^{30,31} At this point there are different proposed chemical pathways leading to the final PDA structure and the polymerization mechanism of dopamine. The proposed mechanism includes intermolecular crosslinking of 5,6 dihydroxyindole or 5,6 indolequinone through branching reactions at positions 2, 3, 4, and 7 finally leads to the eumelanin like polymer polydopamine.³¹ Lee and coworkers suggested that dimers and trimers formed *via* oxidation coupling and then assembled to form the PDA.³² It has also been proposed that PDA is composed entirely of noncovalent assemblies of dopamine, dopamine-quinone and 5,6 dihydroxyindole,³³ whereas Beck and coworkers hypothesised that these molecules polymerize to form a heteropolymer composed of catecholamine, quinone and indole repeat units.³⁴ Possible polymerization mechanisms and structures of polydopamine proposed by researchers is shown in Fig. 2.

Thus, a firmly adherent PDA layer can be formed on the surface of a substrate that is immersed in the dopamine solution for a certain time. The PDA layer can be tightly attached to any kind of substrate by covalent or non-covalent interactions (such as π - π interaction and charge transfer interactions). The PDA coating thickness depends on the concentration of dopamine in solution, deposition time, pH and supplied oxygen in the solution.²⁸ This primary PDA coating can be used without further modification or used as a "primer" onto which a subsequent secondary coating can be applied. The PDA-coated surface layers contain amino groups and phenolic groups, which can react with a variety of molecules *via* Schiff-base and Michael addition chemistries to facilitate immobilize thiol or amine containing molecules and directly reduce metal cations to metal nanoparticles^{22,35} (Fig. 3). Due to great adhesion of PDA with a wide range of polymers, one step PDA coating has been explored as a surface modifier. PDA also serves as a linker for initiators for both atom transfer radical polymerization (ATRP) and reversible addition-fragmentation chain-transfer polymerization (RAFT) techniques (Fig. 3).²⁸ By connecting with these initiators various polymer chains can be grafted onto the PDA modified surfaces *via* surface-initiated ATRP (SI-ATRP) or RAFT. In the last decade, PDA coating has

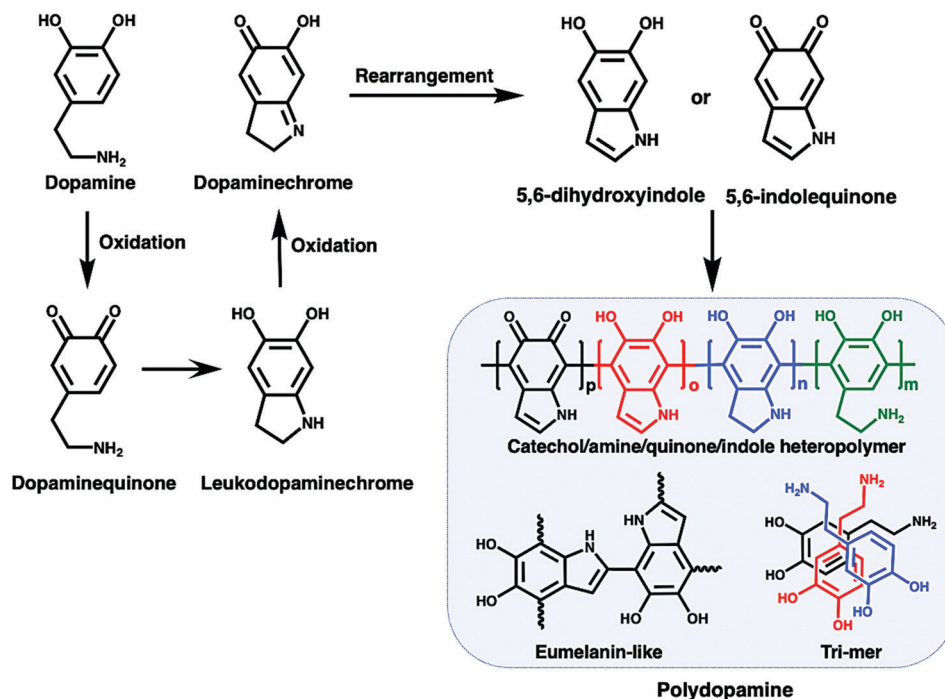


Fig. 2 Possible polymerization mechanism of dopamine. Auto-oxidation of dopamine leads to the formation of dopamine-quinone which undergoes further oxidation and rearrangement to form 5,6-dihydroxyindole and 5,6 indolequinone. Proposed possible structures of final product polydopamine range from non-covalent trimer assemblies, eumelanin-like oligo-indoles to covalent coupling of subunits to yield a catechol/amine/quinone/indole heteropolymer.

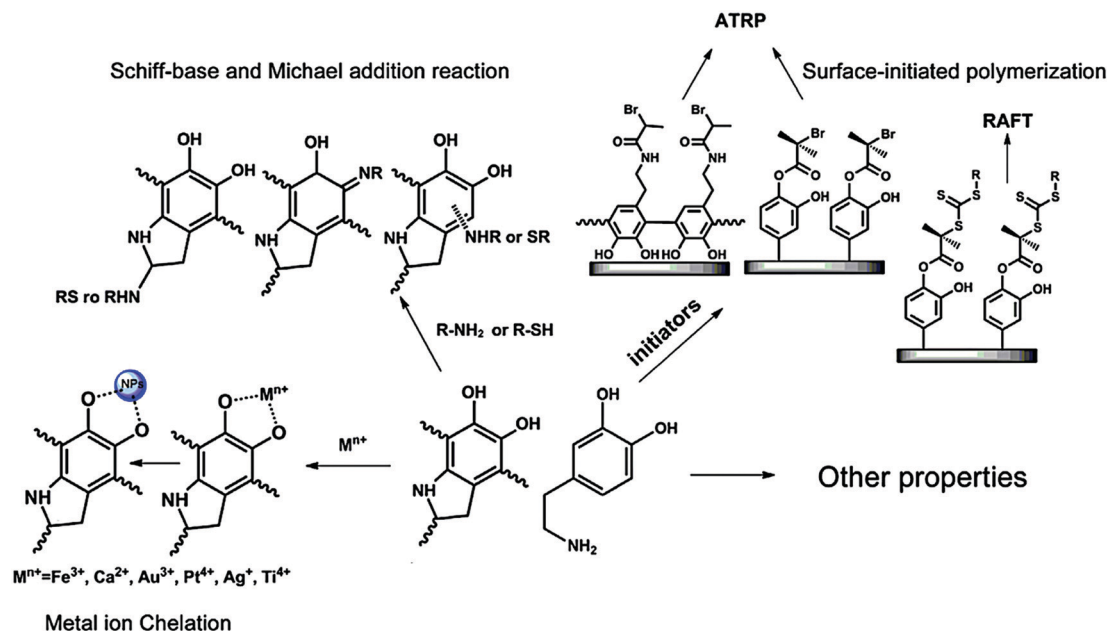


Fig. 3 Chemical properties of dopamine and PDA as representatives of catecholamine and catecholamine-based materials. Reprinted with permission from ref. 28. Copyright 2015 Elsevier.

been utilized in various sectors such as energy storage devices, anti-biofouling, antimicrobial surface, drug delivery, cancer nanomedicine, membrane separation technologies, tissue engineering, artificial spores, mechanically tough hydrogels,

photonic materials, bone regenerations and so on.^{33,36} Among these applications, in most cases PDA coating played the role of an intermediate linker to immobilize other chemical components onto the surface to impart the desired properties.

1.3.2. Comparison of PDA coating with other coating techniques. Prior to the invention of PDA coating, for polymer coating, three major techniques have been utilized in surface modification chemistry such as self-assembled monolayer (SAM),^{37–39} layer-by-layer (LbL) assembly,^{40–42} and plasma treatment.^{43,44} All these methods have specific advantages along with some drawbacks but share a certain surface specificity. Self-assembled monolayers can only be developed on the noble metal (*i.e.* Cu, Pt, Ag, Au) surfaces⁴⁵ using molecules carrying thiol groups through highly specific metal–thiolate bonds³⁸ or can be deposited on the surface of oxides using alkylsilanes⁴⁵ and therefore require matching surface–adsorbate chemistries. On the other hand, robust PDA coating can be deposited onto any kind of surface with secondary reactivity, whether organic or inorganic, whether hydrophilic or hydrophobic; whether blocky, filamentous or granular; and whether metal, polymer or even a living cell.⁴⁶ Another interesting surface functionalization technique is polymer thin film assembly *via* sequential deposition of interacting polymers or layer-by-layer deposition of polyelectrolyte multilayer films of charged surfaces.⁴⁷ Though an LBL assembly is flexible in terms of substrate choice for deposition and allows post modification, it involves multiple deposition steps using interacting polymers and requires modification of polymers for crosslinking or (bio)functionalization.⁴⁸ Therefore, polymers, used in the LBL assembly, need to be synthesized specifically with a view toward LBL use or to provide a new function. In contrast, PDA in its simplest form is a synthesis-free method in which the coating with intrinsic chemical reactivity²⁸ can be built in only one step without the requirement of synthesizing sophisticated polymers or other coating components.³³ As the LBL assembly mostly involves many coating cycles, the formation of multilayers approaching thermodynamic equilibrium is a relatively time-consuming process⁴⁹ and labor intensive. In terms of costs of the instrumentation, production procedures, number and variety of materials that can be assembled, and surfaces to be coated, it generally accepted that polydopamine coating is a more suitable, simple, and versatile surface modification method for the fabrication of robust functional multilayer assemblies.³³

1.4. Dopamine and zwitterion conjugation

Conjugation of zwitterionic polymer with dopamine is a very promising approach to obtain a non-fouling surface where dopamine serves as the surface anchoring motif and the zwitterionic polymer serves as the antifouling motif. Surface coating with a biomimetic zwitterionic polymer has been extensively explored to endow the surface with antifouling properties for medical devices, biosensors and marine coating applications.¹⁰ Because of the high resistance to the non-specific adsorption of proteins and the irreversible adhesion of microorganisms, zwitterionic polymers have been considered as a new generation of antifouling materials.⁵⁰ A polydopamine (PDA) layer on the surface is one of the best candidates as a precursor layer to effectively immobilize a zwitterionic polymer on the surface. Zwitterionic polymers can be immobilized onto

the surface by either a ‘graft to’ or a ‘graft from’ method using dopamine chemistry.^{51,52} The ‘graft to’ method commonly includes synthesis of an end-functionalized polymers and then subsequently adsorbed onto a surface by forming a chemical bond (*i.e.*, chemisorption) or a physical adsorption of polymers *via* specific interactions of the polymers with the surface (*i.e.*, hydrophobic or electrostatic interactions, hydrogen bonds, antibody–antigen, and geometrical complementary match).⁵¹ In ‘graft to’ strategies, polymers may be conveniently characterized prior to grafting but it can be difficult to achieve high grafting density due to steric effects, particularly as the chain length increases. In the case of the ‘graft from’ method, either an initiator is immobilized or generated on a surface and then *in situ* surface-initiated polymerization with monomers is initiated by the immobilized initiators.⁵¹ Compared to the ‘graft to’ method, the “graft from” approach is more promising to synthesizing the polymer brushes on the surface with a high grafting density¹⁰ and well-controlled film thickness. Therefore, dopamine or catechol derivatives can serve as a versatile grafting agent or a ‘primer’ for the convenient and stable anchoring of the zwitterionic polymers onto surfaces by either ‘graft to’ or ‘graft from’ approaches.

Based on the recent studies, for zwitterion and dopamine conjugation, these two approaches can be divided into four more methods which are summarized into Fig. 4. In brief, we have categorized that zwitterionic polymers can be conjugated with dopamine by mostly four methods (Fig. 4): (a) direct modifications by zwitterions with the dopamine group, (b) co-deposition of dopamine with zwitterionic polymers, (c) zwitterionic post modification of PDA coated surface, and (d) surface initiated polymerization of zwitterionic polymers. Here, the first three methods – direct modification, co-deposition and post modification fall under the ‘graft to’ approach and the surface initiated polymerization method mainly refers to the ‘graft from’ approach. The details of each method have been discussed later under Sections 2 and 3.

Researchers have explored extensively both the ‘graft to’ and the ‘graft from’ method using dopamine derivatives to conveniently anchor zwitterionic polymers on the surfaces for achieving ultra-low fouling. Jiang and co-workers have developed a series of efficient surface modification techniques with dopamine derivatives to anchor zwitterionic polymers such as poly(sulfobetaine methacrylate) (PSBMA)^{53–55} and poly(carboxybetaine methacrylate) (PCBMA)^{56–58} to obtain super low fouling surfaces in complex media. Their studies were mostly based on grafting or growing zwitterionic polymer brushes on the surface from a synthesized catecholic atom transfer radical polymerization (ATRP) initiator. Elimelech and coworkers developed PDA assisted dual functional zwitterionic and cationic polymer brush coating *via* activators regenerated by electron transfer atom transfer radical polymerization (ARGET-ATRP) to obtain both antifouling and antibacterial properties.⁵⁹ All these techniques require multiple step treatment to functionalize the surface which would be costly and time consuming. To simplify the surface modification process, Chen *et al.* developed adhesive monomer dopamine

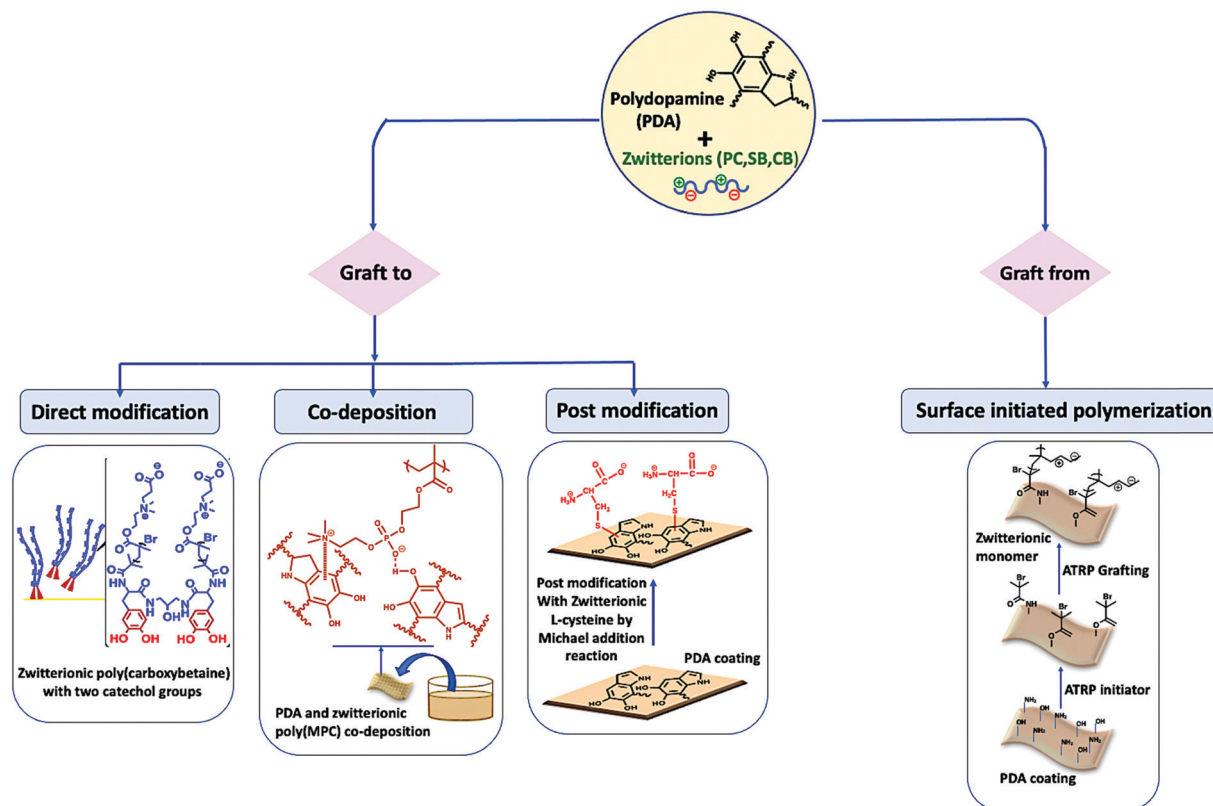


Fig. 4 Categorization of dopamine zwitterion conjugation into 'Graft to' and 'Graft from' approaches. Examples of four different conjugation methods – direct modification, co-deposition post modification and surface initiated polymerization which fall under the two major categorization 'Graft to' and 'Graft from' approaches.

methacrylamide (DMA) containing terpolymers with MPC and cationic monomers by a conventional free radical polymerization and constructed a self-assembled antifouling and contact killing antibacterial surface using a simple one-step dip coating method.⁶⁰ Following this one-step dip coating technique, Xu *et al.* utilized the non-covalent interaction between dopamine and zwitterionic polymers and developed dopamine triggered one step co-deposition of zwitterionic monomers (SBMA, MPC and CMA) to fabricate a non-fouling surface.⁶¹ They were the first to report that dopamine can act as an initiator for the polymerization of acrylate monomers. Previously, they also fabricated antifouling membrane surface and underwater superoleophobic meshes for oil/water separation by poly(SBMA)/PDA one-step codeposition.^{50,62} Emrick *et al.* also utilized this method and co-deposited poly(MPC)/PDA to construct a super hydrophilic antifouling surface using the hydrogen bonding and cation π interpolymer interaction between poly(MPC) and PDA.⁶³ Later, Xu *et al.* improved this co-deposition time by increasing the oxidation rate of dopamine.⁶⁴ As co-deposition rate is completely dependent on the oxidation of dopamine, they added $\text{CuSO}_4/\text{H}_2\text{O}_2$ into the solution to produce reactive oxygen free radicals which accelerated the polymerization and deposition rate of PDA with zwitterionic polymers. As PDA coated layer always contains some reactive catechol or amine groups, thiol-containing zwitterionic polymers can be easily covalently grafted onto PDA coated surface

via click chemistry.^{65,66} Kang,⁶⁵ Lin⁶⁶ and coworkers recently utilized this thiol-ene "click" photopolymerization strategy to graft the polymer brushes of MPC from the PDA coated surface to mitigate the surface fouling. Other researchers also utilized dopamine assisted surface initiator immobilization and subsequent surface initiated polymerization of the zwitterionic monomer to fabricate the antifouling zwitterionic polymer brush.^{67–69}

2. Antifouling strategies with zwitterions and dopamine using 'graft to' approaches

2.1. Direct modification by zwitterions with dopamine group

To achieve a non-fouling surface coating with higher packing density, direct attachment of zwitterionic polymer chains with adhesive groups to the surface in one step is highly desirable because of its simplicity of approach. As the zwitterionic polymer is highly water soluble, it is essential to have a strong surface binding group on the zwitterionic polymer to achieve higher packing density and stability. Catechol containing dopamine is a promising candidate as a surface binding group because of its stable surface anchoring properties on a wide range of surfaces.²² With this in mind, in 2008, Jiang *et al.* first developed a catecholic ATRP initiator (Fig. 5) for grafting zwitterionic polymers on a gold surface to obtain an ultralow

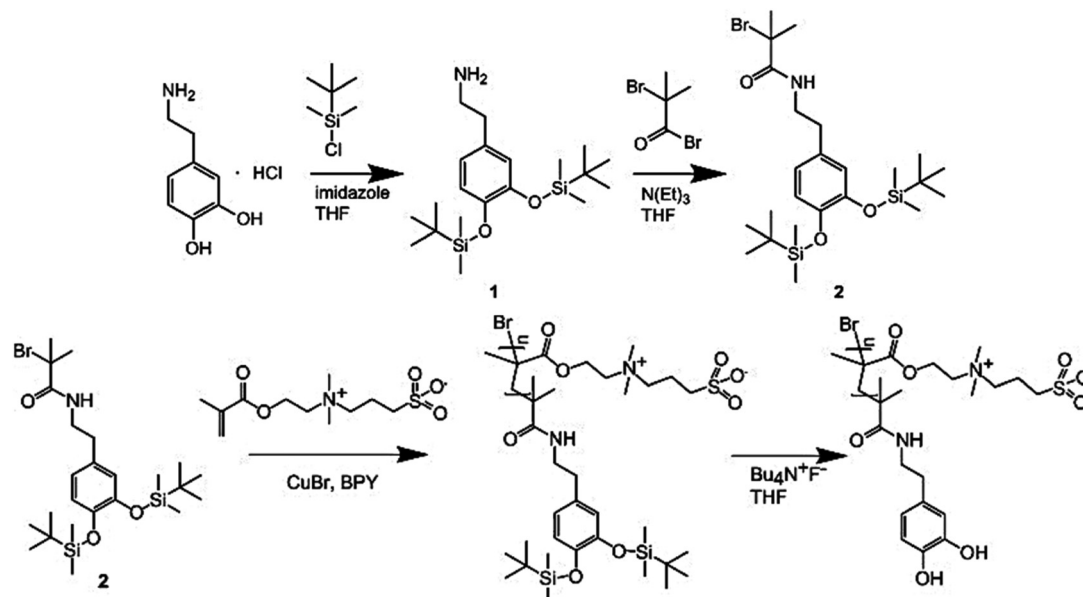


Fig. 5 Reaction steps for the grafting of poly(SBMA) from the catechol initiator via ATRP, followed by the deprotection of hydroxyl groups before surface adhesion. Reprinted with permission from ref. 53. Copyright 2008 Elsevier.

fouling surface in complex media.⁵³ Catecholic oxygens of this initiator were protected by the reaction of dopamine hydrochloride and *t*-butyldimethylsilyl (TBDMS) chloride. With the help of this protected initiator – bromo-2-methyl-*N*-[2-(3,4-di-*t*-butyldimethylsilyloxy)-phenyl] ethyl]propionamide – the polymerization of SBMA was carried out *via* the ATRP method. The catechol containing poly(SBMA) polymer was then grafted onto the surface with a high packing density by simply flowing the polymer solution over the surface and an excellent antifouling surface was achieved.

Jiang *et al.* applied this universal one-step dip-coating method to obtain the antifouling properties of both hydrophilic and hydrophobic surfaces.^{54,70} Zwitterionic sulfobetaine (poly(SBMA)) and extremely hydrophilic carboxybetaine methacrylate (poly(CBMA)) polymers with one catechol chain end from adhesive 3,4-dihydroxyphenyl-L-alanine (DOPA)

(*i.e.* DOPA-poly(SBMA), DOPA-poly(CBMA)) were applied to a variety of surfaces such as hydrophilic metal oxides (silica) and hydrophobic polymer sheets and fibers (nylon, polystyrene, polypropylene, polyvinyl chloride, Teflon *etc.*). To improve the grafting density of catechol containing polymer, Gao *et al.* synthesized two catechol groups containing polymer (pCB₂-catechol₂) with two zwitterionic poly(carboxybetaine) (pCB) arms.⁵⁷ Due to the increased surface anchoring group DOPA in the polymer chain, pCB₂-catechol₂ showed the best non-fouling properties on the grafted surface with strong surface binding and increased surface coverage (Fig. 6).

To increase the coating binding affinity, Sun *et al.* introduced four adhesive L-3,4-dihydroxyphenylalanine (DOPA) groups in the poly(CBMA) polymer chain.⁷¹ The presence of four surface binding groups in one polymer chain (poly(CBMA)-DOPA₄) increased the surface binding affinity along with the

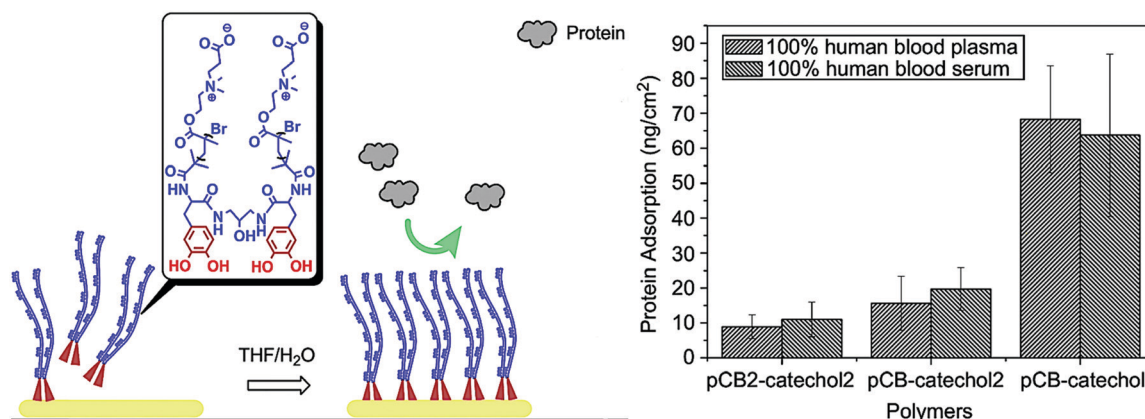


Fig. 6 Schematic diagram of an antifouling Au surface grafted with pCB₂-catechol₂ in THF/H₂O. Reprinted with permission from ref. 57. Copyright 2010 Elsevier.

antifouling properties of coated cellulose paper. Because of the high water solubility of poly(CBMA)-DOPA, it showed weaker binding to hydrophobic surfaces which led to high fouling. To overcome this solubility problem Subdaram *et al.* included dopamine along with poly(CBMA)-DOPA to promote the attachment of this polymer to any kind of surface.⁷⁰ Dopamine can easily polymerize to polydopamine and the catechol group of poly(CBMA)-DOPA polymer chain can react with the amine group present on polydopamine and thus this mixed polymer solution can easily be grafted to any kind of surface with a higher surface coverage. In 2014, Zhao *et al.* developed a series of synthetic bifunctional polymers with catechol conjugation to construct self-cross-linked nanolayers on polymeric substrates *via* the pH induced catechol cross-linking and immobilization.⁵² Several bio-functional polymers, *i.e.*, poly(sodium 4-vinylbenzenesulfonate)-*co*-poly(acrylic acid) (P(SS-*co*-AA)), poly((2-(methacryloyloxy)ethyl) dimethyl-(3-sulfopropyl) ammonium hydroxide)-*co*-poly(acrylic acid) (P(SBMA-*co*-AA)), poly(poly(ethylene glycol) methyl ether methacrylate)-*co*-poly(acrylic acid) (P(EGMA-*co*-AA)), poly(1-vinyl-2-pyrrolidone)-*co*-poly(acrylic acid) (P(VP-*co*-AA)), and poly((2 (methacryloyloxy)ethyl) trimethyl-ammonium chloride)-*co*-poly(acrylic acid) (P(MTAC-*co*-AA)), were first synthesized *via* reversible addition-fragmentation chain transfer (RAFT) polymerization, and then the catecholic molecules (dopamine) were conjugated to these polymers by the facile carbodiimide reaction with acrylic acid units. This mussel adhesion protein inspired dopamine chemistry helped the development of several stable biointerfaces with versatile biological functionalities due to easy immobilization of catechol containing polymers.⁵² Ye *et al.* also utilized this carbodiimide chemistry between hyaluronic acid (HA) and dopamine to make a polyanion of the HA-dopamine conjugate for low fouling surfaces.⁷² Chen *et al.* prepared terpolymer containing dopamine, zwitterionic polymer MPC and cations to construct a recycle antibacterial and antifouling surface.⁶⁰ This terpolymer of dopamine containing dopamine methacrylamide (DMA), cationic 2-(dimethylamino)-ethyl methacrylate (DMAEMA) and MPC was synthesized by conventional free-radical polymerization. Pendent DMA of this terpolymer served as a surface adhesive to successfully coat the polymer solution onto the surface by a simple one-step self-assembly method. Instead of larger polymer chain, Huang *et al.* employed small molecules of zwitterions and dopamine.⁷³ Without any complex polymerization reactions, they conjugated a dopamine molecule with the zwitterionic sulfobetaine moiety to obtain a highly packed adhesive thin film through the pH transition approach for an antifouling biointerface.

2.2. Co-Deposition of dopamine with zwitterionic polymers

Compared to the traditional 'grafting to' method for zwitterionic polymer immobilization, dopamine-assisted co-deposition is more robust, controllable and can be easily adjusted by changing the pH, deposition time, concentration and atmosphere. Dopamine-assisted co-deposition can greatly simplify the procedure of zwitterionic polymer grafting and enrich possible coating functionality by integrating the deposition and functionalization in a single simultaneous step.

In 2012, Lee *et al.* first proposed the concept of dopamine-assisted co-deposition by dissolving dopamine with molecules of a wide range in sizes (10^2 to 10^6 Da) and with various chemistries containing carboxyl, amine, thiol, quaternary ammonium, and/catechol groups.⁷⁴ Later inspired by this dopamine assisted co-deposition concept and to avoid a complicated multistep procedure for grafting zwitterionic polymers, Xu *et al.* first reported one-step dopamine-assisted co-deposition method for constructing antifouling membrane surfaces with poly(SBMA).^{50,62} This one step deposition using a one-pot mixture of dopamine alkaline solution with a polymer, greatly simplifies the coating process and provides an ultrathin composite coating on a variety of surfaces, including glass, silicon, nanofiber, polystyrene, perfluorinated silicon, steel, and microporous polypropylene membranes. Dopamine-assisted co-deposition involves a second component during the deposition of dopamine, including synthetic polymers or monomers, small organic molecules and nanomaterials.

The mechanism of dopamine-assisted deposition involves single or multiple covalent/non-covalent interactions between dopamine and a second component.⁷⁵ Compared to the non-covalent co-deposition, the covalent co-deposition provides a robust network to promote the coating stability. Emrick *et al.* utilized dopamine assisted non-covalent co-deposition technique to fabricate biocompatible, hydrophilic, and fouling-resistant surfaces *via* the co-deposition of dopamine and poly(MPC).^{63,76,77} Both poly(SBMA) and poly(MPC) interact with PDA through non-covalent linkages, including electrostatic interaction and hydrogen bonding. Poly(SBMA) interacts with PDA through local electrostatic interactions between deprotonated phenol groups of dopamine and the quaternary ammonium of poly(SBMA).⁶² Poly(MPC) interacts with PDA through phenol-phospholipid hydrogen bonding and cation- π interactions (Fig. 7).⁶³

Coating conditions, deposition time, oxidation of dopamine and solution concentration control the coating thickness of PDA assisted coating. Oxygen diffusion barrier was still a problem for oxidative dopamine polymerization which has been further solved by Xu *et al.*^{64,78} They accelerated the

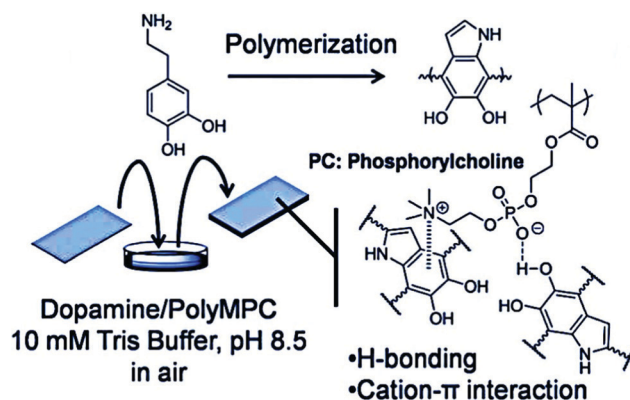


Fig. 7 Schematic illustration of PDA and poly(MPC) co-deposition coating and the mechanism of the PDA-poly(MPC) non-covalent interactions. Reprinted with permission from ref. 63. Copyright 2016 Wiley Online.

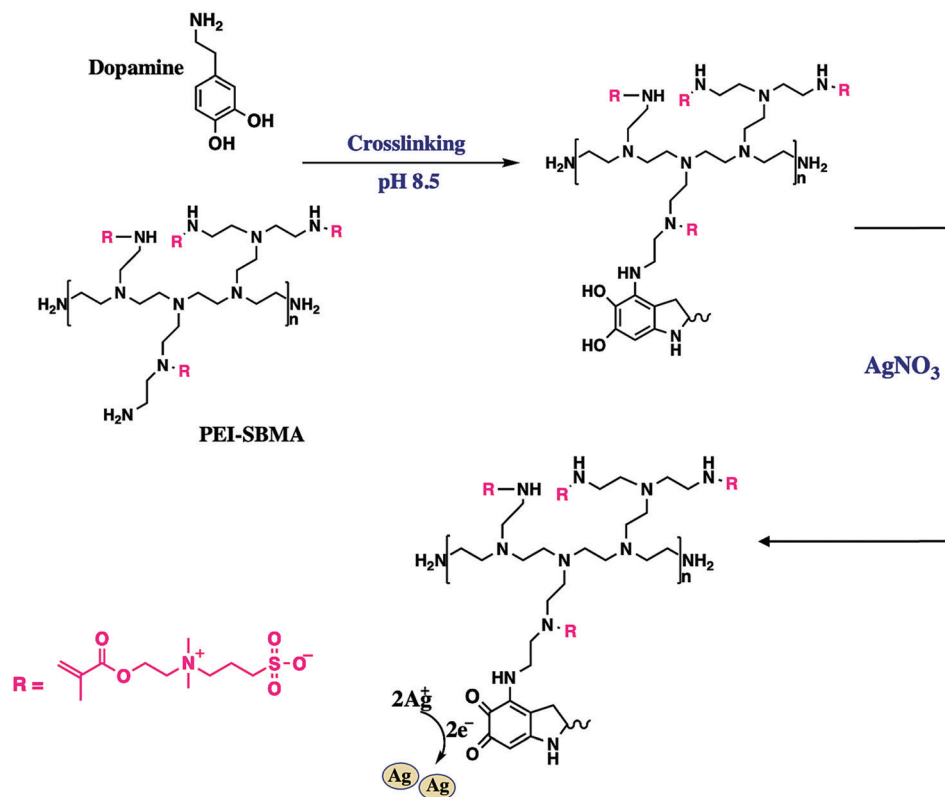


Fig. 8 Schematic presentation of PDA/PEI-SBMA co-deposition and AgNP incorporation. Redrawn with permission from ref. 78. Copyright 2016 Wiley Online.

polymerization of dopamine and the deposition rate of PDA coatings by using $\text{CuSO}_4/\text{H}_2\text{O}_2$ as a trigger. Cu^{2+} and H_2O_2 produce a great deal of reactive oxygen free radicals in an alkaline medium which play a key role in rapid polymerization of dopamine and improve the deposition rate of PDA coatings. In this way, they reduced the PDA based co-deposition time from 18 h to 1 h only. PDA based co-deposited coatings can also be used for secondary functionalization of the surface because of the presence of some unreacted catechol group in PDA. Unreacted $-\text{OH}$ groups of PDA based coatings can be utilized to incorporate metal nanoparticles into the coating which can impart some antibacterial activities as well. Zhao *et al.* recently reported a novel antibacterial and antifouling membrane *via* co-deposition of PDA and PEI-SBMA, followed by incorporating bactericidal silver nanoparticles (AgNPs).⁷⁹ One catechol group of PDA can lose two electrons and reduce the diffused Ag^+ ion to Ag^0 and transformed itself into stable 1,2-benzoquinone as shown in Fig. 8. Thus, the catechol group of PDA can reduce metal ions, including silver and gold ions, into metallic nanoparticles without any reducing agent.

2.3. Zwitterionic post modification of PDA coated surface

The catechol motif of polydopamine proves to be more beneficial when it comes to post modification strategies, especially in the context of conjugation with thiol or amine containing molecules. To be specific, the oxidized quinone form of catechol of PDA can undergo reactions with nucleophilic amine or

thiol groups *via* Michael addition or Schiff-base reaction (Fig. 9) to form a covalently grafted functional polymer layer.⁸⁰ In a Michael addition reaction between amine terminated polymer and polydopamine, an amine is attached to the catechol ring at the 4- or 5-position, and in a Schiff base reaction the amine attacks the 2-position of the catechol ring and an imine is created.⁸¹ This reaction mechanism between the oxidized quinone of PDA and the amine containing polymer depends on the type of amine. Generally, aromatic amines favor the Michael-addition reaction, whereas aliphatic amines favor the Schiff base reaction.⁸¹ Similarly, the electrophilic oxidized quinone of PDA is highly reactive towards nucleophilic thiol groups and can react with the thiol terminated polymer *via* the Michael addition reaction to form catechol-thiol adducts.

2.3.1. PDA and thiol group containing zwitterionic polymer conjugation. Catechol-thiol adducts have been detected in many different systems. Inspired by this dopamine chemistry, in 2013, Jian Shen *et al.* covalently grafted thiol and amino groups containing zwitterionic material cysteine to the PDA coated PET sheets *via* thiol-ene Michael addition reaction to improve the antifouling and hemocompatibility properties.⁸² Being pH responsive, the post modified surface with cysteine coating can act as a self-cleaning surface which was further investigated by Shevate *et al.* by anchoring cysteine on the PDA coated membranes *via* Michael-addition reaction at 50 °C to alleviate the antifouling ability of the membrane with foulant solution.⁸³ Cui *et al.* also immobilized zwitterionic glutamic

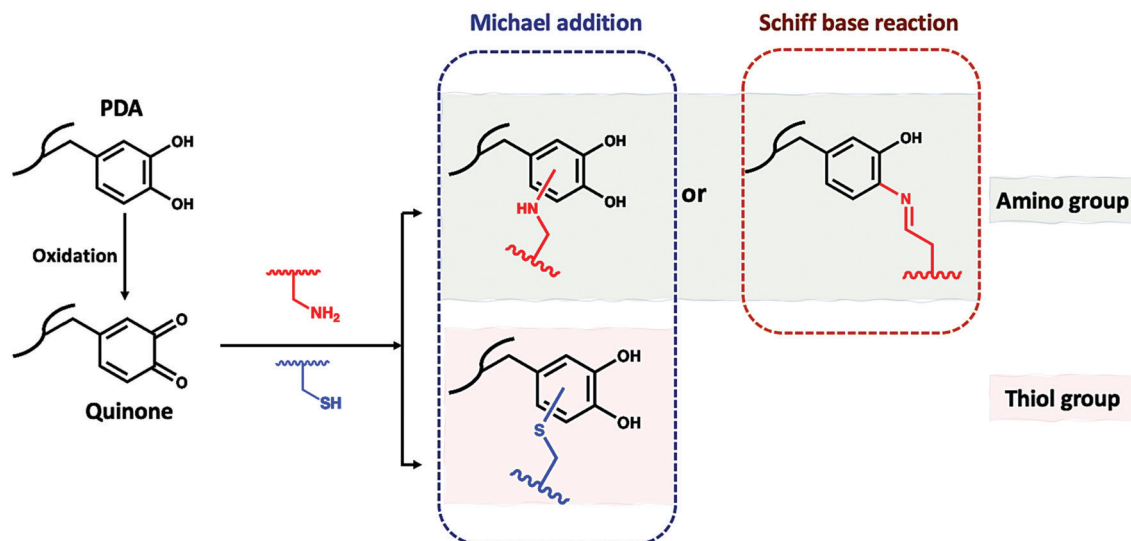


Fig. 9 Reactions of oxidized quinone of PDA with amines and thiols via Michael addition and Schiff base reaction. In a Michael addition reaction both the amino group and the thiol group attacks the catechol ring at the 4- or 5-position, whereas in a Schiff base reaction the amino group attacks the 2-position of the catechol ring of PDA.

acid and lysine-based peptides onto PDA coated noble metals, metal oxides, polymers and semiconductors *via* Michael-addition reaction between thiol of peptides and quinone of PDA groups to obtain surface-independent low fouling surfaces.⁸⁴ For long-term application and to overcome the high water solubility of zwitterionic polymer this two-step PDA assisted coating technique is convenient and effective. To prove this concept, Lin *et al.* grafted dithiol-containing poly(MPC-co-2-(methacryloyloxy)ethyl lipoate (MAEL)) copolymer to PDA coated *via* Michael addition reaction.^{44,66} They first modified the membrane with a thin layer of PDA coating in Tris-buffer solution. Then covalently grafted the dithiol containing poly(MPC-co-MAEL) copolymer synthesized using the RAFT polymerization technique. The thiol-functional group containing zwitterionic copolymers grafting showed increased hydrophilicity and superior long term antifouling properties.⁶⁶ Xu *et al.* utilized click chemistry to assemble zwitterionic and cationic binary polymer brushes onto PDA-anchored stainless steel (SS) surfaces.⁶⁵ Zwitterionic poly(MPC) was first polymerized grafting from the PDA functionalized SS surface *via* thiol-ene “click” reaction. Alkynyl-modified cationic poly(2-(methacryloyloxy) ethyl trimethylammonium chloride) (alkynyl-poly(META)) was then grafted to the surface *via* azide-alkyne “click” reaction. This poly(MPC)/poly(META) binary polymer brushes grafted *via* click chemistry to the PDA modified surface, showed very stable (more than 30 days) and durable antifouling and antibacterial properties.⁶⁵

2.3.2. PDA and amine group containing zwitterionic polymer conjugation. Similar to these ‘thiol-functional’ groups containing zwitterionic polymer grafting, the PDA layer also can act as an intermediate layer to immobilize ‘amine-containing’ molecules *via* Michael addition or Schiff-base reaction. Shen *et al.* successfully immobilized small zwitterionic molecule lysine (contains two amine –NH₂ groups) to the

pre-coated PDA layer *via* amino–ene Michael addition reaction and obtained an excellent biocompatible and antifouling surface.⁸⁵ To avoid the selectivity of only the amine functional group containing a zwitterionic molecule grafting on PDA coated surface, Wang *et al.* synthesized a copolymer of acrylate zwitterionic molecule SBMA with 2-aminoethyl methacrylate (AEMA) named as poly(SBMA-co-AEMA).⁸⁶ The amino groups (–NH₂) of the AEMA segment reacted with PDA-modified surface by Michael addition or Schiff base reactions and opened a new possibility of grafting zwitterionic copolymers with an amino group pendant chain. Besides, by controlling the content of –NH₂ in the copolymers through adjusting the ratio of Zwitterion/AEMA the interaction of the copolymer with PDA coating can also be enhanced.

Instead of utilizing the amine functional group from zwitterionic molecule/polymer, researchers had also employed the amine groups in the bulk PDA layer and grafted amine free zwitterionic molecule/polymer on the PDA coated surface covalently. Using this two-step post modification approach, Liu *et al.* developed a new conjugation technique for the formation of β -amino carbonyl linkages between the amine functional group of PDA and acrylate/acrylamide zwitterionic molecules (Sulfobetaine acrylamide (SBAA) and SBMA) *via* the aza-Michael reaction (Fig. 10) for developing the multifunctional antifouling interfaces in a substrate-independent fashion.⁸⁷ The aza-Michael addition is of synthetic importance, in which the nucleophilic amine groups of PDA coating attach to the α,β -unsaturated carbonyl compounds.

To enhance the interaction between PDA coating and zwitterionic compound, by utilizing amine groups of the PDA coated surface, recently Narain and coworkers have developed a robust strategy by covalently grafting adhesive monomer DMA and zwitterionic MPC containing copolymer poly(MPC-co-DMA) onto an amino (–NH₂) rich polyethylenimine (PEI)/PDA

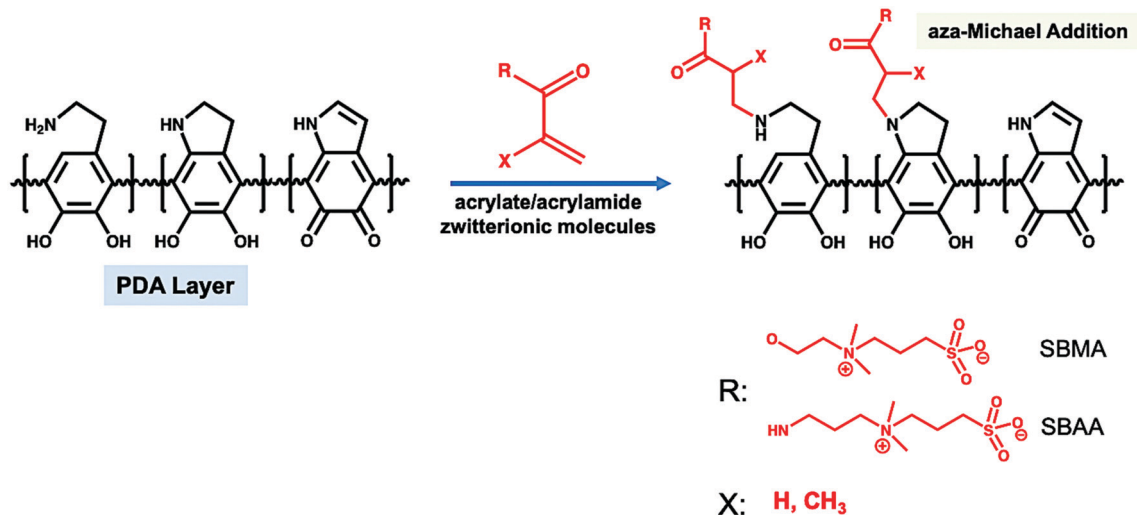


Fig. 10 PDA and acrylate/acrylamide zwitterionic molecule conjugation by the formation of β -amino carbonyl linkages via aza-Michael addition reaction. Redrawn with permission from ref. 87. Copyright 2016 American Chemical Society.

co-deposited surface.⁸⁸ Co-deposition of PEI and PDA helped to decorate the surface with more anchoring sites (more nucleophilic amine groups on the surface) available for zwitterionic compound grafting. The resulting surface was also treated for *in situ* deposition of antimicrobial silver nanoparticles (AgNPs), facilitated by the presence of catechol groups of the coating to impart antibacterial properties.

3. Antifouling strategies with zwitterions and dopamine using the 'graft from' approach

3.1. Surface initiated polymerization of zwitterionic polymers

One of the most effective methods to prevent biofouling is grafting the zwitterionic monomers from a surface *via* surface-initiated atom transfer radical polymerization (SI-ATRP). Due to its controlled/living feature, this SI-ATRP has been widely developed to generate non-fouling polymer brushes with high packing densities and controllable film thicknesses.⁸⁹ One of the challenges of the conventional SI-ATRP method is the selectivity of chemically reactive surfaces for covalent immobilization of ATRP initiators.⁹⁰ The most conventionally used ATRP initiators that can assemble on a substrate surface are thiol, chlorosilane and alkoxy-silane-based initiators.^{91–93} The binding chemistries of these ATRP initiators are substrate-dependent. For example, the thiol-based initiator is only effective on gold and some noble metal substrates, and the chlorosilane or alkoxy-silane-based initiators are effective on oxides surfaces.⁹⁴

To overcome this limitation of SI-ATRP for the selective substrate surface, mussel-inspired robust adhesive dopamine chemistry has been extensively studied to deposit the PDA layer onto a wide range of surfaces and thereby acts as an efficient platform for further surface immobilization of ATRP

initiators.^{67,95–97} To show the efficiency of dopamine for the immobilization ATRP initiators, Chang *et al.* compared catecholic dopamine and organosilane as the respective anchor site for subsequent immobilization of ATRP initiator onto a stainless steel surface and poly(SBMA) polymerization.⁶⁸ In their study, poly(SBMA) brushes grown from the respective dopamine and silane-assembly layers were compared for the effectiveness in resisting the adhesion of plasma protein, blood cells, mammalian cells, and bacteria. Results showed that poly(SBMA) grafted from polydopamine interfacial layers achieved better bioadhesion resistance than from the silane-based assembly layer. Such a dopamine-initiated ATRP can be applied to nearly any substrate to achieve the desired surface chemical characteristics, shapes and sizes for growing antifouling polymer brushes from the surface. This dopamine-initiated ATRP method can be divided into two approaches. In the first approach, most of the researchers first coated the surface with dopamine followed by immersing the PDA coated substrate into the ATRP initiator aqueous solution for cross-linking the PDA oligomers with the ATRP initiator and then grew zwitterionic polymer brushes from the modified surface^{95–98} (Fig. 11). The amount of initiators immobilized on surfaces determines the graft ratio of zwitterionic polymers on membrane surfaces.⁵⁶ Usually in SI-ATRP, the initiator's primary concentration on a surface remains very low compared to that used for bulk or solution ATRP.⁹⁹ Therefore, to obtain thicker polymer brushes at a faster growth rate and to establish an equilibrium between dormant and active chains during SI-ATRP, an excess amount of deactivating Cu(II) complex in the form of CuBr is usually added.

In another approach, the researchers studied modified polydopamine as an initiator layer for SI-ATRP, which was synthesized by reacting dopamine with an ATRP initiator before deposition.^{67,100,101} This ATRP initiator modified dopamine film subsequently act as the initiator to initiate polymerization of antifouling monomers from the surface using activators

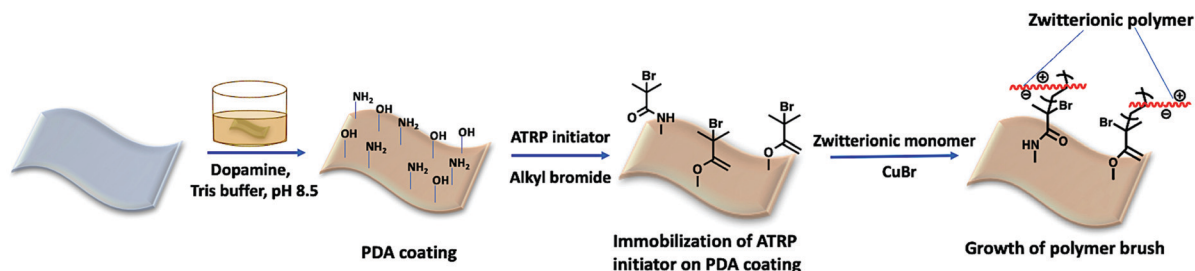


Fig. 11 Schematic illustration for surface-initiated ATRP for generating zwitterionic polymer brushes on the PDA coated surface through immobilization of the ATRP initiator onto the PDA coated surface.

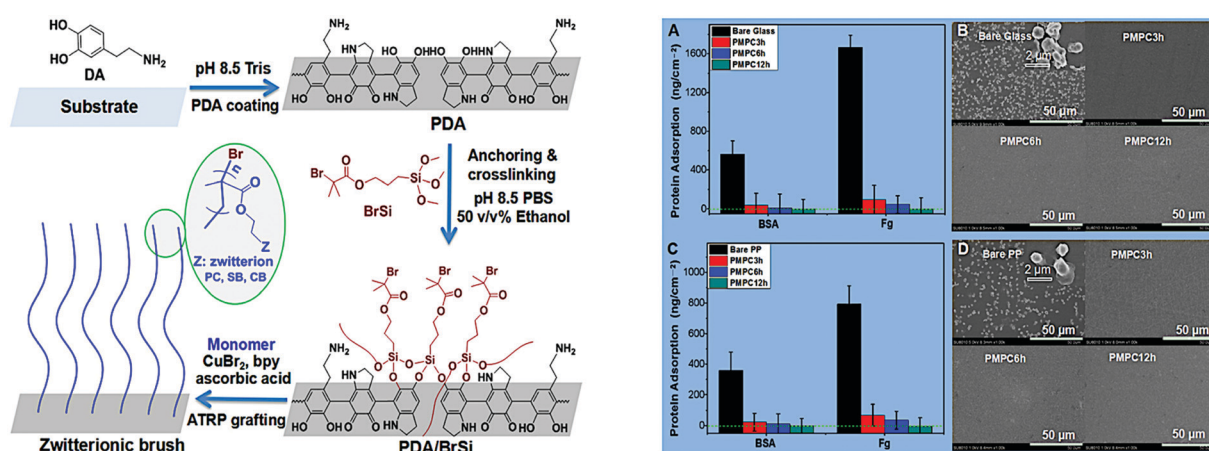


Fig. 12 Schematic illustration of zwitterionic polymer brush construction on a material-independent substrate by PDA universal adhesion, trimethoxysilyl group cross-link, and ATRP initiator covalent immobilization (left), (A and C) protein adsorption and (B and D) platelet adhesion results on (A and B) glass and (C and D) polypropylene (PP) surfaces grafted with PMPC brushes for different hours. The graft polymerization was carried out in aqueous solution containing 20 mg mL⁻¹ MPC, 0.0036 mg mL⁻¹ CuBr₂, 0.0152 mg mL⁻¹ bipyridine (bpy), and 0.14 mg mL⁻¹ ascorbic acid at 50 °C. Protein adsorption amount was measured by the BCA assay (right). Reprinted with permission from ref. 90. Copyright 2020 American Chemical Society.

regenerated by electron transfer (ARGET) – ATRP. As the conventional SI-ATRP has few limitations including the use of relatively high concentration of catalyst complexes, rigorous removal of oxygen and the purification of polymers, the newly developed ATRP initiating system ARGET-ATRP can help to mitigate these limitations by introducing a reducing agent (*i.e.*, ascorbic acid) and reducing the concentration of the catalyst.^{90,98} Recently, Gong *et al.* reported an universal strategy of preparing stable zwitterionic polymer brushes by integrating the advantages of both PDA chemistry and ARGET-ATRP.⁹⁰ In their study, a PDA adhesive layer was first dip-coated on a substrate, followed by covalent immobilization of 3-trimethoxysilyl propyl 2-bromo-2-methylpropionate (SiBr, ATRP initiator) on the PDA coated surface *via* a condensation reaction between the silicon hydroxyl and the PDA hydroxyl groups. Then, SI-ARGET-ATRP was performed in a zwitterionic monomer solution catalyzed by the parts per million level of CuBr₂ without deoxygenation (Fig. 12). This conveniently fabricated zwitterionic polymer brush coatings were demonstrated to have stable, ultralow fouling, and extremely blood compatibility.

4. Applications

In the past few years, polydopamine and zwitterion conjugation has aroused considerable interest for various types of applications ranging from marine to biomedical industries. Because of the unique molecular structure of zwitterionic compounds, they have been widely used as promising candidate to tune the interface properties of materials in various fields. In conjunction with adhesive domain dopamine or dopamine derivatives, these zwitterionic compounds are broadly utilized in biomedical applications due to their dual capabilities for fouling resistance and functionalization. In most of the applications polydopamine based simple and versatile surface modification techniques play three common roles: surface coatings, interlayers for better compatibility and interlayers for further functionalization. Apart from antifouling applications, this dopamine-zwitterion novel conjugation has been employed in wastewater treatment, as drug carriers, in wound healing and in biosensors. In Table 1, we have summarized the recent advances in the dopamine and dopamine derivatives mediated surfaces in conjugation with zwitterions for various applications.

Table 1 Summary of the recent applications of dopamine–zwitterion conjugation

Substrate	Antifouling zwitterionic compound	Conjugation mechanism with dopamine/dopamine derivatives	Applications	Ref.
Stainless steel	SBMA	ATRP	General anti-bio adhesive against plasma protein, blood cells, mammalian cells, and bacteria.	68
Commercial thin-film composite (TFC) membranes	SBMA	ARGET-ATRP	Mitigate biofouling on TFC membranes used in desalination technology for global water production	59
Cellulose nanofibers	MPC	Sequential deposition and co deposition	Antifouling nanofiber mats for tissue engineering scaffolds and water purification technologies.	76
—	1,3-Propane sultone	Ring opening reaction	Scarless wound healing	102
Electrospun poly(L-lactic acid) (PLLA) film	SBMA	ATRP	Ultra low fouling in complex media for biomedical application	55
Sensor chip of surface plasma resonance (SPR) instrument.	MPC	Amidation reaction with intermediate PDA layer	Advanced antifouling surfaces for clinical catheters, vascular stent, artificial lung, hemodialyzer, biomedical devices and long circulating nanocarriers	103
Polydimethylsiloxane (PDMS) membranes	CBMA	ATRP	Anticoagulant biomaterials with reduced platelet fouling	58
Silicon wafer, glass and stainless steel mesh	MPC, SBMA	Hydrogen bonding, cation– π interaction and electrostatic interaction	Underwater super oleophobic surface for oil/water separation	62 and 63
Cellulose paper, gold and silica	CBMA	ATRP	Functionalized biosensor for sensing and detection in complex media	57, 71 and 104
Titanium surface	MPC	ATRP	For improved hemocompatibility of cardiovascular stent and some other biomaterials.	95
Stainless steel	MPC	SI-ATRP	Marine antifouling	96
Polypropylene	SBMA	SI-ATRP	Oriented antibody immobilization and immunoassay	105
Poly(ethylene terephthalate) (PET)	Cysteine, Lysine	Michael addition reaction	Hemocompatible anti-biofouling	82 and 85
Ultrafiltration (UF) membranes	SBMA	Michael addition reaction	Wastewater reuse in water treatment	106
Polyethersulfone ultrafiltration membrane	SBMA	Schiff-base and Michael addition	Ultrafiltration membranes for reduced biofouling in skim milk filtration	107
PVDF membrane	MPC	Amidation reaction	Oil/water separation	108
Poly(lactic acid) membrane	SBMA	ATRP	Hemodiafiltration for hemodialysis	109
Poly(lactic-co-glycolic acid) nanoparticles	CBMA	Schiff-base	Nano-drug carriers	23
Mesoporous silica nanoparticles	CBMA	ATRP	Stealth multifunctional biocarriers for drug delivery/ diagnostics	110
Glass, mica, and gold	SBMA	Crosslinking	Antifouling, anti-freezing and antifogging	111

5. Summary and outlook

In this review, we have summarized recent studies on four major surface modification techniques using the dopamine chemistry and zwitterion conjugation to mitigate biofouling. Most of these strategies have shown excellent results in combating the problem of fouling but some of them are also associated with shortcomings related to stability, and complexity of the method of fabrication.

Superhydrophilic zwitterionic polymers have been extensively used as promising antifouling materials for surface modification. However, it remains a challenge to form a thick layer with high zwitterion content on the surfaces with different degrees of hydrophilicity. Since the zwitterionic polymer brush chain non-covalently attached to the surface often suffers stability problems, research studies have a widely exploited mussel-inspired polydopamine (PDA) adhesive functional layer to attach the zwitterion layer stably *via* thiol and amine chemistry. Due to material-independent coating ability, simplicity of the deposition process, and the unique and broad ranging

capabilities for donating sufficient binding sites for antifouling agents PDA coating has become the most convenient and universal coating material to graft other polymers. The conjugation of PDA and the zwitterionic polymer acts as one of the most effective antifouling systems with durable performance. The key to grafting zwitterionic polymers using dopamine is to optimize the functional groups of the zwitterionic monomers that can react with PDA to achieve uniform layers and high zwitterion content on the surface for maximum antifouling properties. Despite several notable achievements in PDA–zwitterion conjugation, there are still some challenges in this field including better understanding of PDA formation mechanisms, enhancement of the mechanical robustness of the PDA coating, and extension of the PDA coating approach for large-scale industrial applications. With more insight into the polymerization mechanisms of dopamine and related materials, new rationally designed materials with high functionality can be achieved, addressing the need for advanced antifouling materials for long-term applications in biomedical, agriculture, food industry and other sectors.

Although in recent years significant achievements have been made in the rational design and fabrication of antifouling surfaces using dopamine chemistry, the majority of the proposed chemistries between dopamine and other molecules are still at the preliminary stage of exploration. The underlying theories and mechanisms of developing zwitterionic brush from PDA coated surfaces are still not clear. The polydopamine formation and zwitterionic polymer brush grafting and post-functionalization still need to be optimized. There should be more elaborative exploration on the proposed chemistries for antifouling surface development using dopamine and zwitterion conjugation. An obvious gap between fundamental research and industrial application with several major challenges need to be addressed such as the scale-up difficulty, high capital costs, uncertain durability and chemical stability in practical long-time operations. Finally, it should be emphasized that ultimate goal of anti-fouling surface construction is to achieve a high and durable performance to better meet the requirements of industrial or large scale applications.

Conflicts of interest

There are no conflicts to declare.

References

- 1 J. Park, G. A. Gill, J. E. Strivens, L. J. Kuo, R. T. Jeters, A. Avila, J. R. Wood, N. J. Schlafer, C. J. Janke, E. A. Miller, M. Thomas, R. S. Addleman and G. T. Bonheyo, *Ind. Eng. Chem. Res.*, 2016, **55**, 4328–4338.
- 2 J. L. Harding and M. M. Reynolds, *Trends Biotechnol.*, 2014, **32**, 140–146.
- 3 E. Lindner, *Biofouling*, 1992, **6**, 193–205.
- 4 G. D. Bixler and B. Bhushan, *Philos. Trans. R. Soc., A*, 2012, **370**, 2381–2417.
- 5 Q. Yu, L. K. Ista and G. P. López, *Nanoscale*, 2014, **6**, 4750–4757.
- 6 C. M. Magin, S. P. Cooper and A. B. Brennan, *Mater. Today*, 2010, **13**, 36–44.
- 7 G. A. O'Toole, *Nature*, 2002, **416**, 695–696.
- 8 D. M. Yebra, S. Kiil and K. Dam-Johansen, *Prog. Org. Coat.*, 2004, **50**, 75–104.
- 9 V. B. Damodaran and S. N. Murthy, *Biomater. Res.*, 2016, **20**, 1–11.
- 10 M. He, K. Gao, L. Zhou, Z. Jiao, M. Wu, J. Cao, X. You, Z. Cai, Y. Su and Z. Jiang, *Acta Biomater.*, 2016, **40**, 142–152.
- 11 S. Chen, L. Li, C. Zhao and J. Zheng, *Polymer*, 2010, **51**, 5283–5293.
- 12 J. Zheng, L. Li, H. K. Tsao, Y. J. Sheng, S. Chen and S. Jiang, *Biophys. J.*, 2005, **89**, 158–166.
- 13 J. Zhao, Q. Shi, S. Luan, L. Song, H. Yang, H. Shi, J. Jin, X. Li, J. Yin and P. Stagnaro, *J. Membr. Sci.*, 2011, **369**, 5–12.
- 14 Q. Yu, Z. Wu and H. Chen, *Acta Biomater.*, 2015, **16**, 1–13.
- 15 Y. H. Zhao, X. Y. Zhu, K. H. Wee and R. Bai, *J. Phys. Chem. B*, 2010, **114**, 2422–2429.
- 16 N. Shahkaramipour, T. N. Tran, S. Ramanan and H. Lin, *Membranes*, 2017, **7**.
- 17 S. Jiang and Z. Cao, *Adv. Mater.*, 2010, **22**, 920–932.
- 18 M. Bernards and Y. He, *J. Biomater. Sci., Polym. Ed.*, 2014, **25**, 1479–1488.
- 19 L. Zhang, M. Tang, J. Zhang, P. Zhang, J. Zhang, L. Deng, C. Lin and A. Dong, *J. Appl. Polym. Sci.*, 2017, **134**, 44632.
- 20 J. F. Jhong, A. Venault, L. Liu, J. Zheng, S. H. Chen, A. Higuchi, J. Huang and Y. Chang, *ACS Appl. Mater. Interfaces*, 2014, **6**, 9858–9870.
- 21 J. B. Schlenoff, *Langmuir*, 2014, **30**, 9625–9636.
- 22 H. Lee, S. M. Dellatore, W. M. Miller and P. B. Messersmith, *Science*, 2007, **318**, 426–430.
- 23 J. Park, T. F. Brust, H. J. Lee, S. C. Lee, V. J. Watts and Y. Yeo, *ACS Nano*, 2014, **8**, 3347–3356.
- 24 Y. Chen, X. Feng, Y. Zhao, X. Zhao and X. Zhang, *Mol. Pharm.*, 2020, **17**, 1215–1225.
- 25 J. H. Waite and M. L. Tanzer, *Science*, 1981, **212**, 1038–1040.
- 26 J. H. Waite and X. Qin, *Biochemistry*, 2001, **40**, 2887–2893.
- 27 H. G. Silverman and F. F. Roberto, *Mar. Biotechnol.*, 2007, **9**, 661–681.
- 28 H. C. Yang, J. Luo, Y. Lv, P. Shen and Z. K. Xu, *J. Membr. Sci.*, 2015, **483**, 42–59.
- 29 H. A. Lee, Y. Ma, F. Zhou, S. Hong and H. Lee, *Acc. Chem. Res.*, 2019, **52**, 704–713.
- 30 J. Jiang, L. Zhu, L. Zhu, B. Zhu and Y. Xu, *Langmuir*, 2011, **27**, 14180–14187.
- 31 F. Bernsmann, V. Ball, F. Addiego, A. Ponche, M. Michel, J. J. D. A. Gracio, V. Toniazio and D. Ruch, *Langmuir*, 2011, **27**, 2819–2825.
- 32 S. Hong, Y. S. Na, S. Choi, I. T. Song, W. Y. Kim and H. Lee, *Adv. Funct. Mater.*, 2012, **22**, 4711–4717.
- 33 J. H. Ryu, P. B. Messersmith and H. Lee, *ACS Appl. Mater. Interfaces*, 2018, **10**, 7523–7540.
- 34 J. Liebscher, R. Mrówczyński, H. A. Scheidt, C. Filip, N. D. Haidade, R. Turcu, A. Bende and S. Beck, *Langmuir*, 2013, **29**, 10539–10548.
- 35 Y. Zeng, W. Liu, Z. Wang, S. Singamaneni and R. Wang, *Langmuir*, 2018, **34**, 4036–4042.
- 36 K. S. Schanze, H. Lee and P. B. Messersmith, *ACS Appl. Mater. Interfaces*, 2018, **10**, 7521–7522.
- 37 R. G. Nuzzo and D. L. Allara, *J. Am. Chem. Soc.*, 1983, **105**, 4481–4483.
- 38 J. C. Love, L. A. Estroff, J. K. Kriebel, R. G. Nuzzo and G. M. Whitesides, *Chem. Rev.*, 2005, **105**, 1103–1170.
- 39 A. Ulman, *Chem. Rev.*, 1996, **96**, 1533–1554.
- 40 J. J. Kirkland, *Anal. Chem.*, 1965, **37**, 1458–1461.
- 41 R. K. Iler, *J. Colloid Interface Sci.*, 1966, **21**, 569–594.
- 42 J. J. Richardson, J. Cui, M. Björnholm, J. A. Braunger, H. Ejima and F. Caruso, *Chem. Rev.*, 2016, **116**, 14828–14867.
- 43 E. M. Liston, L. Martinu and M. R. Wertheimer, *J. Adhes. Sci. Technol.*, 1993, **7**, 1091–1127.
- 44 P. K. Chu, J. Y. Chen, L. P. Wang and N. Huang, *Mater. Sci. Eng., R*, 2002, **143**–206.

- 45 V. Ball, D. Del Frari, M. Michel, M. J. Buehler, V. Toniazzo, M. K. Singh, J. Gracio and D. Ruch, *Bionanoscience*, 2012, **2**, 16–34.
- 46 X. Tan, P. Gao, Y. Li, P. Qi, J. Liu, R. Shen, L. Wang, N. Huang, K. Xiong, W. Tian and Q. Tu, *Bioact. Mater.*, 2021, **6**, 285–296.
- 47 J. Borges and J. F. Mano, *Chem. Rev.*, 2014, **114**, 8883–8942.
- 48 M. E. Lynge, R. Van Der Westen, A. Postma and B. Städler, *Nanoscale*, 2011, **3**, 4916–4928.
- 49 S. Zhao, F. Caruso, L. Dahne, G. Decher, B. G. De Geest, J. Fan, N. Feliu, Y. Gogotsi, P. T. Hammond, M. C. Hersam, A. Khademhosseini, N. Kotov, S. Leporatti, Y. Li, F. Lisdat, L. M. Liz-Marzan, S. Moya, P. Mulvaney, A. L. Rogach, S. Roy, D. G. Shchukin, A. G. Skirtach, M. M. Stevens, G. B. Sukhorukov, P. S. Weiss, Z. Yue, D. Zhu and W. J. Parak, *ACS Nano*, 2019, **13**, 6151–6169.
- 50 R. Zhou, P. F. Ren, H. C. Yang and Z. K. Xu, *J. Membr. Sci.*, 2014, **466**, 18–25.
- 51 B. Y. Yu, J. Zheng, Y. Chang, M. C. Sin, C. H. Chang, A. Higuchi and Y. M. Sun, *Langmuir*, 2014, **30**, 7502–7512.
- 52 X. Liu, J. Deng, L. Ma, C. Cheng, C. Nie, C. He and C. Zhao, *Langmuir*, 2014, **30**, 14905–14915.
- 53 G. Li, G. Cheng, H. Xue, S. Chen, F. Zhang and S. Jiang, *Biomaterials*, 2008, **29**, 4592–4597.
- 54 H. S. Sundaram, X. Han, A. K. Nowinski, J. R. Ella-Menye, C. Wimbish, P. Marek, K. Senecal and S. Jiang, *ACS Appl. Mater. Interfaces*, 2014, **6**, 6664–6671.
- 55 W. Yang, H. S. Sundaram, J. R. Ella, N. He and S. Jiang, *Acta Biomater.*, 2016, **40**, 92–99.
- 56 G. Cheng, G. Li, H. Xue, S. Chen, J. D. Bryers and S. Jiang, *Biomaterials*, 2009, **30**, 5234–5240.
- 57 C. Gao, G. Li, H. Xue, W. Yang, F. Zhang and S. Jiang, *Biomaterials*, 2010, **31**, 1486–1492.
- 58 K. A. Amoako, H. S. Sundaram, A. Suhaib, S. Jiang and K. E. Cook, *Adv. Mater. Interfaces*, 2016, **3**, 1500646.
- 59 G. Ye, J. Lee, F. Perreault and M. Elimelech, *ACS Appl. Mater. Interfaces*, 2015, **7**, 23069–23079.
- 60 B. L. Wang, T. W. Jin, Y. M. Han, C. H. Shen, Q. Li, Q. K. Lin and H. Chen, *J. Mater. Chem. B*, 2015, **3**, 5501–5510.
- 61 C. Zhang, M. Q. Ma, T. T. Chen, H. Zhang, D. F. Hu, B. H. Wu, J. Ji and Z. K. Xu, *ACS Appl. Mater. Interfaces*, 2017, **9**, 34356–34366.
- 62 P. F. Ren, H. C. Yang, Y. N. Jin, H. Q. Liang, L. S. Wan and Z. K. Xu, *RSC Adv.*, 2015, **5**, 47592–47598.
- 63 C. C. Chang, K. W. Kolewe, Y. Li, I. Kosif, B. D. Freeman, K. R. Carter, J. D. Schiffman and T. Emrick, *Adv. Mater. Interfaces*, 2016, **3**, 1500521.
- 64 C. Zhang, H. N. Li, Y. Du, M. Q. Ma and Z. K. Xu, *Langmuir*, 2017, **33**, 1210–1216.
- 65 G. Xu, P. Liu, D. Pranantyo, L. Xu, K. G. Neoh and E. T. Kang, *Ind. Eng. Chem. Res.*, 2017, **56**, 14479–14488.
- 66 N. Shahkaramipour, C. K. Lai, S. R. Venna, H. Sun, C. Cheng and H. Lin, *Ind. Eng. Chem. Res.*, 2018, **57**, 2336–2345.
- 67 J. Kuang and P. B. Messersmith, *Langmuir*, 2012, **28**, 7258–7266.
- 68 M. C. Sin, Y. M. Sun and Y. Chang, *ACS Appl. Mater. Interfaces*, 2014, **6**, 861–873.
- 69 C. Liu, J. Lee, J. Ma and M. Elimelech, *Environ. Sci. Technol.*, 2017, **51**, 2161–2169.
- 70 H. S. Sundaram, X. Han, A. K. Nowinski, N. D. Brault, Y. Li, J. R. Ella-Menye, K. A. Amoaka, K. E. Cook, P. Marek, K. Senecal and S. Jiang, *Adv. Mater. Interfaces*, 2014, **1**, 1400071.
- 71 F. Sun, K. Wu, H. C. Hung, P. Zhang, X. Che, J. Smith, X. Lin, B. Li, P. Jain, Q. Yu and S. Jiang, *Anal. Chem.*, 2017, **89**, 10999–11004.
- 72 H. Ye, Y. Xia, Z. Liu, R. Huang, R. Su, W. Qi, L. Wang and Z. He, *J. Mater. Chem. B*, 2016, **4**, 4084–4091.
- 73 C. J. Huang, L. C. Wang, J. J. Shyue and Y. C. Chang, *Langmuir*, 2014, **30**, 12638–12646.
- 74 S. M. Kang, N. S. Hwang, J. Yeom, S. Y. Park, P. B. Messersmith, I. S. Choi, R. Langer, D. G. Anderson and H. Lee, *Adv. Funct. Mater.*, 2012, **22**, 2949–2955.
- 75 W. Z. Qiu, H. C. Yang and Z. K. Xu, *Adv. Colloid Interface Sci.*, 2018, **256**, 111–125.
- 76 K. W. Kolewe, K. M. Dobosz, K. A. Rieger, C. C. Chang, T. Emrick and J. D. Schiffman, *ACS Appl. Mater. Interfaces*, 2016, **8**, 27585–27593.
- 77 A. Y. Kirschner, C. C. Chang, S. Kasemset, T. Emrick and B. D. Freeman, *J. Membr. Sci.*, 2017, **541**, 300–311.
- 78 C. Zhang, Y. Ou, W. X. Lei, L. S. Wan, J. Ji and Z. K. Xu, *Angew. Chem., Int. Ed.*, 2016, **55**, 3054–3057.
- 79 Y. Xie, C. Tang, Z. Wang, Y. Xu, W. Zhao, S. Sun and C. Zhao, *J. Mater. Chem. B*, 2017, **5**, 7186–7193.
- 80 Y. H. Ding, M. Floren and W. Tan, *Biosurf. Biotribol.*, 2016, **2**, 121–136.
- 81 J. Yang, M. A. Cohen Stuart and M. Kamperman, *Chem. Soc. Rev.*, 2014, **43**, 8271–8298.
- 82 P. Li, X. Cai, D. Wang, S. Chen, J. Yuan, L. Li and J. Shen, *Colloids Surf., B*, 2013, **110**, 327–332.
- 83 R. Shevate, M. Kumar, M. Karunakaran, M. N. Hedhili and K. V. Peinemann, *J. Membr. Sci.*, 2017, **529**, 185–194.
- 84 J. Cui, Y. Ju, K. Liang, H. Ejima, S. Lörcher, K. T. Gause, J. J. Richardson and F. Caruso, *Soft Matter*, 2014, **10**, 2656–2663.
- 85 X. Zhi, P. Li, X. Gan, W. Zhang, T. Shen, J. Yuan and J. Shen, *J. Biomater. Sci., Polym. Ed.*, 2014, **25**, 1619–1628.
- 86 L. Chen, L. Tan, S. Liu, L. Bai and Y. Wang, *J. Biomater. Sci., Polym. Ed.*, 2014, **25**, 766–785.
- 87 C. Y. Liu and C. J. Huang, *Langmuir*, 2016, **32**, 5019–5028.
- 88 A. B. Asha, Y. Chen, H. Zhang, S. Ghaemi, K. Ishihara, Y. Liu and R. Narain, *Langmuir*, 2019, **35**, 1621–1630.
- 89 T. Gillich, E. M. Benetti, E. Rakhmatullina, R. Konradi, W. Li, A. Zhang, A. D. Schlüter and M. Textor, *J. Am. Chem. Soc.*, 2011, **133**, 10940–10950.
- 90 N. Li, T. Li, X. Y. Qiao, R. Li, Y. Yao and Y. K. Gong, *ACS Appl. Mater. Interfaces*, 2020, **12**, 12337–12344.
- 91 H. Ma, J. Hyun, P. Stiller and A. Chilkoti, *Adv. Mater.*, 2004, **16**, 338–341.
- 92 K. Matyjaszewski, P. J. Miller, N. Shukla, B. Immaraporn, A. Gelman, B. B. Luokala, T. M. Siclován, G. Kickelbick,

- T. Valiant, H. Hoffmann and T. Pakula, *Macromolecules*, 1999, **32**, 8716–8724.
- 93 X. He, W. Yang and X. Pei, *Macromolecules*, 2008, **41**, 4615–4621.
- 94 S. Edmondson, C. D. Vo, S. P. Armes and G. F. Unali, *Macromolecules*, 2007, **40**, 5271–5278.
- 95 W. Ma, P. Yang, J. Li, S. Li, P. Li, Y. Zhao and N. Huang, *Appl. Surf. Sci.*, 2015, **349**, 445–451.
- 96 J. Wang and J. Wei, *Appl. Surf. Sci.*, 2016, **382**, 202–216.
- 97 J. Jiang, P. Zhang, L. Zhu, B. Zhu and Y. Xu, *J. Mater. Chem. B*, 2015, **3**, 7698–7706.
- 98 X. Jin, J. Yuan and J. Shen, *Colloids Surf., B*, 2016, **145**, 275–284.
- 99 W. C. Wang, J. Wang, Y. Liao, L. Zhang, B. Cao, G. Song and X. She, *J. Appl. Polym. Sci.*, 2010, **117**, 534–541.
- 100 G. Li, H. Xue, G. Cheng, S. Chen, F. Zhang and S. Jiang, *J. Phys. Chem. B*, 2008, **112**, 15269–15274.
- 101 M. Ginic-Markovic, T. Barclay, K. T. Constantopoulos, T. Al-Ghamdi, A. Blok, E. Markovic and A. V. Ellis, *RSC Adv.*, 2015, **5**, 63017–63024.
- 102 B. Peng, X. Lai, L. Chen, X. Lin, C. Sun, L. Liu, S. Qi, Y. Chen and K. W. Leong, *ACS Omega*, 2017, **2**, 6053–6062.
- 103 C. M. Xing, F. N. Meng, M. Quan, K. Ding, Y. Dang and Y. K. Gong, *Acta Biomater.*, 2017, **59**, 129–138.
- 104 N. D. Brault, C. Gao, H. Xue, M. Piliarik, J. Homola, S. Jiang and Q. Yu, *Biosens. Bioelectron.*, 2010, **25**, 2276–2282.
- 105 J. Zhao, R. Mo, L. M. Tian, L. J. Song, S. F. Luan, J. H. Yin and L. Q. Ren, *Chin. J. Polym. Sci.*, 2018, **36**, 472–478.
- 106 N. Shahkaramipour, S. N. Ramanan, D. Fister, E. Park, S. R. Venna, H. Sun, C. Cheng and H. Lin, *Ind. Eng. Chem. Res.*, 2017, **56**, 9202–9212.
- 107 T. G. Barclay, H. M. Hegab, A. Michelmore, M. Weeks and M. Ginic-Markovic, *J. Ind. Eng. Chem.*, 2018, **61**, 356–367.
- 108 F. N. Meng, M. Q. Zhang, K. Ding, T. Zhang and Y. K. Gong, *J. Mater. Chem. A*, 2018, **6**, 3231–3241.
- 109 L. J. Zhu, F. Liu, X. M. Yu, A. L. Gao and L. X. Xue, *J. Membr. Sci.*, 2015, **475**, 469–479.
- 110 Y. Zhu, H. S. Sundaram, S. Liu, L. Zhang, X. Xu, Q. Yu, J. Xu and S. Jiang, *Biomacromolecules*, 2014, **15**, 1845–1851.
- 111 M. Vatankhah-Varnosfaderani, X. Hu, Q. Li, H. Adelnia, M. Ina and S. S. Sheiko, *ACS Appl. Mater. Interfaces*, 2018, **10**, 20869–20875.